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Can football teach a robot to move?

This article was written by Roberto Shu, senior staff research engineer, and Yeuhi Abe, manager - controls & reinforcement learning.

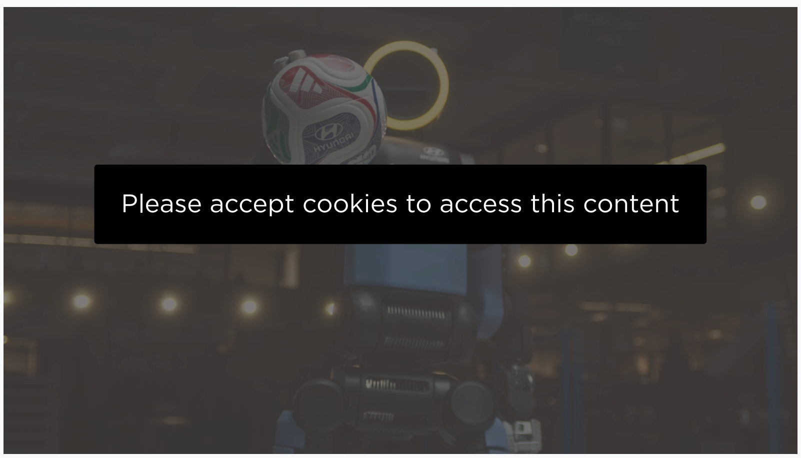
Boston Dynamics makes incredible robots, but we're bu … g
creative, and having fun. With the FIFA World Cup 2026
try something new. The School of Football is a part of H
learning football and mastering a complex Ghost Rabon
celebrate the ways that football brings people together … ed
to bring that energy into our lab and see if football can

watching back game tape, analyzing their performance, and adjusting their training routines. We applied the same logic to our Atlas® humanoid robot, using human reference and reinforcement learning to train Atlas to perform a previously unseen trick shot.

The Ghost Rabona

Our goal was to showcase the incredible mobility, agility, and whole-body control enabled by our very capable humanoid robot. We came up with the Ghost Rabona, combining a fake out step over with a crossed-leg rabona kick.

- Step 1: Atlas walks towards the ball.
- Step 2: Atlas fakes the shot with its left leg.
- Step 3: Atlas crosses its right leg behind to strike the ball.



This move is complicated for a human to perform, much less a robot. Atlas needs to move fast for a convincing fake. It needs power and agility to fully take off from the ground and land again, while still remaining balanced to complete the kick. This combination of different skills pushes the limits of physical intelligence. Making a robot move dynamically and powerfully is exactly the kind of research problem we love to solve at Boston Dynamics.

Learning from (Human) D

Just as a human athlete must practice to master intricat ... s.
We use reinforcement learning to teach the robot. Atlas ... v
to handle the physics of its own body to mimic human r

The first step is to create a reference input from human
programming, turning what used to be a complex codi ... can
come from various sources—including video, teleoperation, and motion capture. For this particular task, we

For the Ghost Rabona, we worked with a football player to develop and record a super dynamic motion. I (Roberto) also took a turn in the motion capture suit for some of the drills and basic kicks, so a lot of the motions that you see is me performing those motions and then training a robot to replicate my behaviors.

0:04

Human to Robot Motion Pipeline

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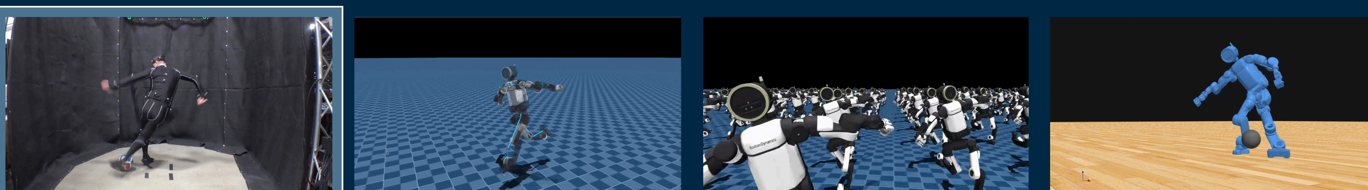

1. Human Motion Capture
0:04

2. Motion Retargeting
0:06

3. RL Training
0:20

4. Sim Evaluation
0:08

Using human motion capture allows for the rapid collec… m with a distinctively "human" style. The difficulty lies in a… isn't a direct correspondence between human and robo… kinematic structure is fundamentally unique.Through a … as kinematics.

performed by the human.

Atlas practices in simulation, with thousands of simulations in parallel happening in GPUs in the cloud. And over time the system learns the actions necessary to realize the behavior in simulation. It's similar to how humans learn through trial and error, except the robot gets to try things much faster than a human could: the equivalent of a full year's worth of physical trial and error in the span of just 24 hours.

Once Atlas learns to perform the skill in simulation we deploy on the hardware. For almost all of Atlas' skills, the learned policy worked first try on the robot. If something fails we go back to our training, make adjustments, and improve the behavior.

Transferrable Skills

Every time Atlas moves into a new environment, there's going to be new training—similar to any new hire learning the ins and outs of this specific job—where it can operate, what it needs to do, how it needs to interact with the world. While the task is not the same, many of the same tools we use to train the robot for football transfer to training the robot to do a job in a [warehouse] or in a [factory].

There's also transferable skills we can hone playing football that translate into useful tasks. Unlike other sports that separate locomotion from manipulation, football demands mastering both at once. Athletes must balance and run while precisely controlling a ball with their feet—a significant hurdle for humanoid robots. Success requires a whole-body controller that dynamically coordinates every joint as a single system, allowing the robot to strike the ball powerfully while maintaining balance. This same whole-body coordination is the essence of what Atlas needs to perform manipulation tasks in the workplace.

More broadly, training generalist behavior for robots requires a vast variety of data and training. We also train Atlas on more mundane, practical applications that it will do in real work environments, but thinking—and training—outside of the box is one of the keys to unlocking generalizable behaviors.

While Atlas was engineered for industrial and factory applications, its capabilities extend far beyond those environments. The demonstrations featured here illustra
These skills are just the beginning of what Atlas can do.
World Cup™ and we're excited to show the Next of robo

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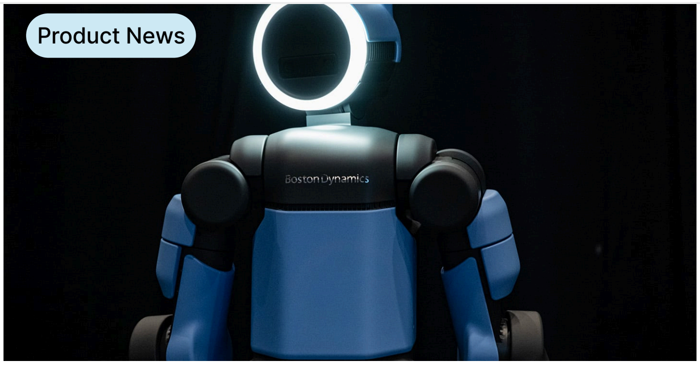

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